
1 Tree shape
Yule
γ
I_C
N
50
100
200
2 Traits
u optimum
σ breadth
h abundance
u σ h E
S0 S1 S2 S2σ S2h S3a S3b S3σ
H0 H1
phylogenetic signal
λ = 0.5
none
generating E
independent E
3 Assembly
A static
E
B dynamic
1000
time steps
a
a sorting
mass effects
a none
phylogenetic
4 archetypes
4 Analysis
W × Q = P
E ~ D_P
PERMANOVA
Mantel
PCPS–GLM
RDA
PCC
site shuffle
taxa shuffle
max(p)
p_{site} p_{taxa}
999 permutations
5 Inference
α
type I error
under H_0
signal
power
under H_1
γ
I_C
×1000
replicates
per cell